

# The Effect of Boarding Strategies on Aircraft Boarding Efficiency

A Simulation-Based Study with an Exact Longest-Path (Max-Plus) Theory | Author: Zihao Huang | October 2026

## Abstract

This study compares three aircraft boarding strategies — *random*, *back-to-front*, and *zoning* (5 zones) — using a single-aisle simulation. Contrary to intuition, **random boarding is the fastest and back-to-front is the slowest**, and the gap widens with stow time and cabin size. We derive an **exact theoretical model** that computes the boarding time directly (without tick-by-tick simulation) as the length of a **longest path** in a max-plus recurrence; it reproduces the simulation **bit-for-bit** (verified over >25,000 cases), and yields provable lower bounds. The theory — corroborated by measured mechanism metrics (parallelism and blocking) — explains *why* clustered boarding is slow: consecutive same-row passengers stack stowing weights on a single aisle cell and lengthen the critical path.

## 1. Background and Research Question

Boarding time affects on-time performance and operating cost. Airlines often use "zone boarding" or "back-to-front boarding", but whether these beat random boarding is debated. We test:

**Hypothesis H:** Organized strategies (back-to-front / zoning) board faster than random boarding.

## 2. Methods

### 2.1 Simulation model

- Single-aisle cabin; passengers enter from the front and walk to their row.
- On reaching the row, a passenger stows luggage, occupying the aisle for **S** ticks; followers cannot pass.
- One cell of forward motion per tick; no overtaking; one passenger enters per tick.
- Total ticks until all are seated = boarding time (smaller = faster).

### 2.2 Variables

| Type               | Variable                                                        |
|--------------------|-----------------------------------------------------------------|
| Independent        | Boarding strategy (Random / Back-to-front / Zones-5)            |
| Dependent          | Total boarding time $T$ (ticks)                                 |
| Control (baseline) | Random boarding (normalized to 1.00)                            |
| Controlled         | Rows $R$ ; seats per row (6); entry rate (1/tick); repeats (20) |

### 2.3 Mechanism metrics

The simulator records, each tick, the number of passengers **stowing simultaneously** (parallelism) and the number of **blocking events** (walkers stuck behind a stower).

## 2.4 Theoretical model: an exact longest-path (max-plus) formulation

We model boarding as a **discrete-time single-lane exclusion process** and derive an **exact recurrence** for  $T$ . Let

- cells be  $1..R$ ; passenger  $i$  ( $i = 1..N$ ) enters in order;  $r_i$  = target row;  $S$  = stow time;
- $P_i^c$  = the tick at which passenger  $i$  reaches cell  $c$ ;
- $j = j_i(c)$  = the last passenger before  $i$  that reaches cell  $c$ .

Passenger  $i$  reaches cell  $c$  at the **later** of two times: its own pace, and the moment cell  $c$  becomes free:

$$P_i^c = \max(P_i^{c-1} + 1, B_i^c), \text{ where } B_i^c = \{ P_j^c + 1 + S \text{ if } r_j = c \text{ (j stows at row } c); P_j^{c+1} \text{ if } r_j > c \text{ (j passes through)} \}$$

with the exact entrance conventions (entering is its own tick; right after entering one may move an extra cell):

$$P_i^1 = \max(i, B_i^1), \quad P_i^2 = \max(P_i^1, B_i^2), \quad P_i^c = \max(P_i^{c-1} + 1, B_i^c) \text{ for } c \geq 3.$$

At the entrance cell, when the passenger ahead only passes through ( $r_j > 1$ ), entering is its own step, so  $B_i^1 = P_j^2 + 1$ .

The finish time of passenger  $i$  and the total time are

$$F_i = P_i^{r_i} + [r_i \geq 2] + S - 1, \quad T = \max_i F_i.$$

**Interpretation.** Every  $P_i^c$  is a *max of two terms* ("my own pace" vs "waiting for the person ahead"), so  $T$  is the length of the **longest chain of waits** — a **longest path** (last-passage percolation) value on the space–time lattice.

### 2.4.1 Worked example

Take  $R = 3$  rows, one seat per row,  $S = 1$ , and boarding order  $r = [3, 1, 2]$  (passenger 1 goes to row 3, passenger 2 to row 1, passenger 3 to row 2). Applying the recurrence cell by cell:

| Passenger $i$ | $r_i$ | Cell-by-cell computation                                                                                 | Finish $F_i$        |
|---------------|-------|----------------------------------------------------------------------------------------------------------|---------------------|
| 1             | 3     | $P_1 = \max(1, 0) = 1; P_2 = \max(1, 0) = 1; P_3 = \max(1+1, 0) = 2$                                     | $2 + 1 + 1 - 1 = 3$ |
| 2             | 1     | $B_1 = P_1^2 + 1 = 2; P_1 = \max(2, 2) = 2$                                                              | $2 + 0 + 1 - 1 = 2$ |
| 3             | 2     | $B_1 = P_2^1 + S = 3 \rightarrow P_1 = \max(3, 3) = 3; B_2 = P_1^3 = 2 \rightarrow P_2 = \max(3, 2) = 3$ | $3 + 1 + 1 - 1 = 4$ |

Thus  $T = \max(3, 2, 4) = 4$ , which matches the simulation exactly.

## 2.5 Lower bounds

**Proposition 1 (entrance bound).**  $T \geq N$ . Proof: at most one passenger enters per tick, and all  $N$  must have entered.

**Proposition 2 (occupancy bound).**  $T \geq (\sum_i r_i + N \cdot S) / R$ . Proof: passenger  $i$  occupies  $r_i$  cell-ticks while walking and  $S$  while stowing, so total occupancy is  $\sum_i r_i + N \cdot S$ ; at most  $R$  cells can be occupied in any tick. For 6 seats per row ( $\sum_i r_i = 3R(R+1)$ ):  **$T \geq 3(R+1) + 6S$ .**

## 2.6 Data collection

Strategies with randomness (Random, Zones) are run **20 times and averaged**; Back-to-front is deterministic. Results are reported relative to random (baseline = 1.00).

## 3. Results

### 3.1 Main result ( $R = 30$ rows, $S = 3$ )

| Strategy      | Average boarding time (ticks) | Relative to random |
|---------------|-------------------------------|--------------------|
| Random        | 331.9                         | 1.00 (baseline)    |
| Zones (5)     | 402.6                         | 1.21               |
| Back-to-front | 627                           | 1.89               |

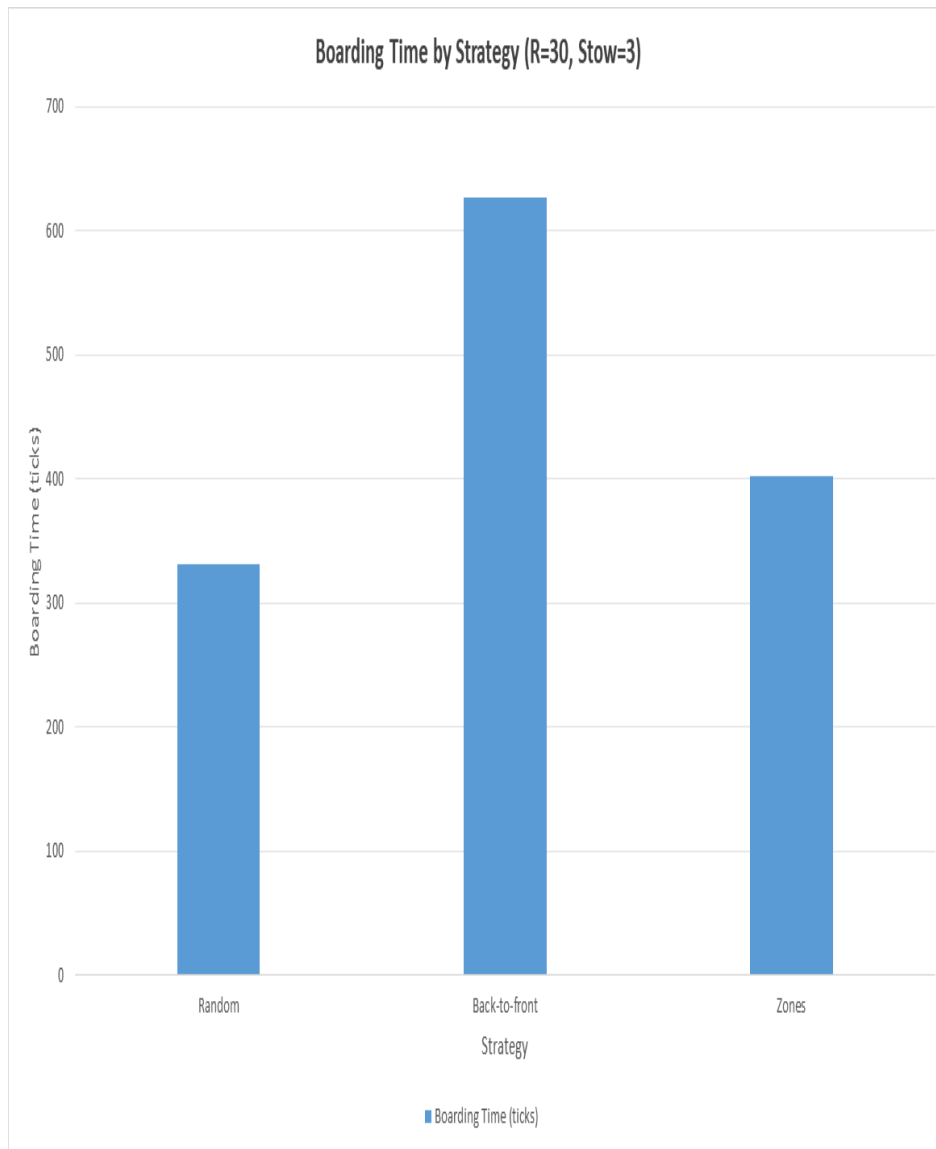


Figure 1. Boarding time by strategy (R=30, S=3): random fastest, back-to-front slowest.

### 3.2 Sensitivity analysis

#### (a) Stow time (fixed R = 30).

| S | Random | Back-to-front | Zones (5) | Rel. random (back) | Rel. random (zones) |
|---|--------|---------------|-----------|--------------------|---------------------|
| 1 | 238.9  | 325           | 249.9     | 1.36               | 1.05                |
| 3 | 331.9  | 627           | 402.6     | 1.89               | 1.21                |
| 6 | 471.5  | 1080          | 633.8     | 2.29               | 1.34                |

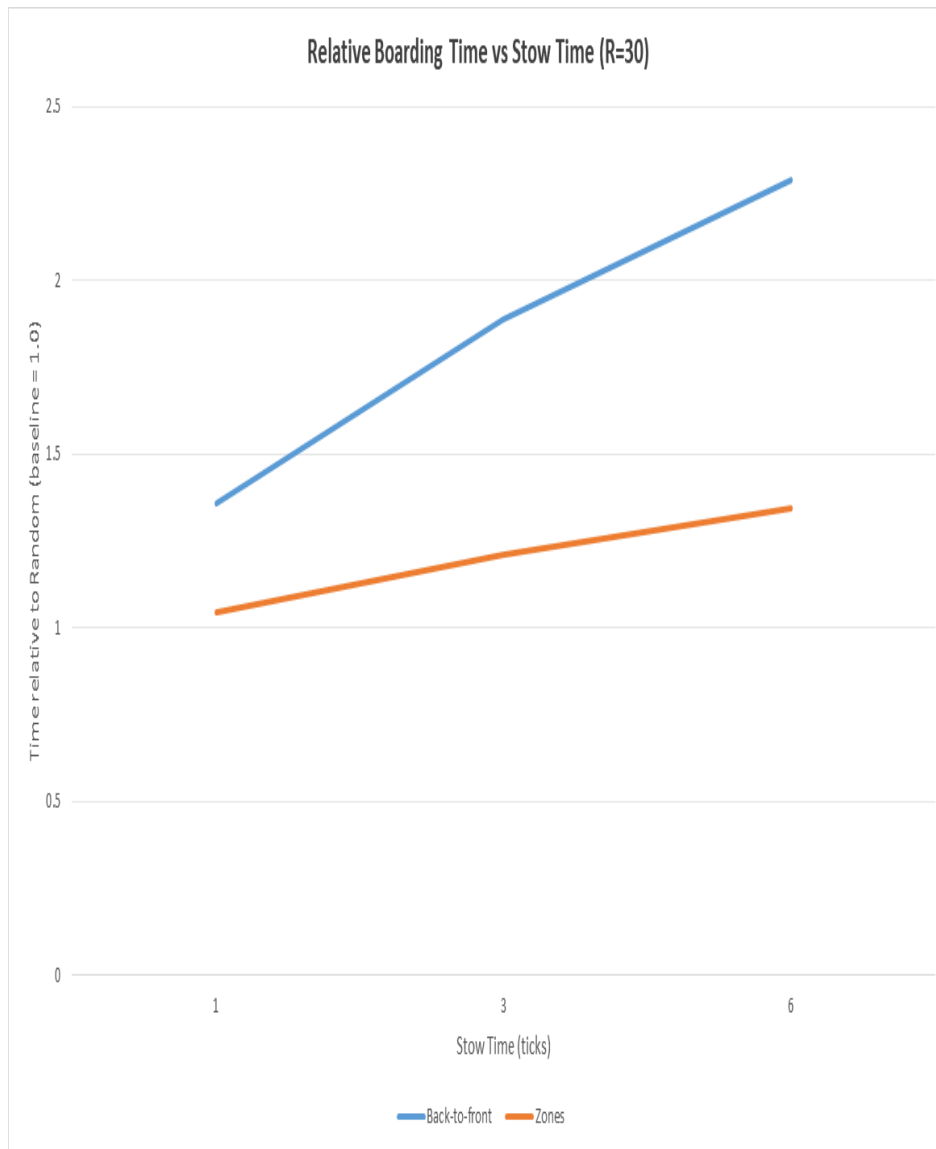


Figure 2. Relative boarding time vs stow time (R=30).

**(b) Number of rows (fixed S = 3).**

| Rows R | Random | Back-to-front | Zones (5) | Rel. random (back) | Rel. random (zones) |
|--------|--------|---------------|-----------|--------------------|---------------------|
| 20     | 234.3  | 417           | 288.7     | 1.78               | 1.23                |
| 30     | 331.9  | 627           | 402.6     | 1.89               | 1.21                |
| 40     | 423.6  | 837           | 507.3     | 1.98               | 1.20                |

**3.3 Why: the empirical mechanism and its exact theory**

**Empirical mechanism.** The measured mechanism agrees with the theory: the fastest strategy is also the most parallel. The average number of passengers stowing at the same moment is **1.09** for random versus **0.57** for back-to-front, and the total number of blocking events is **~950** versus **6,525** (Figure 3) — random lets many rows stow simultaneously, while back-to-front forces one-by-one stowing.

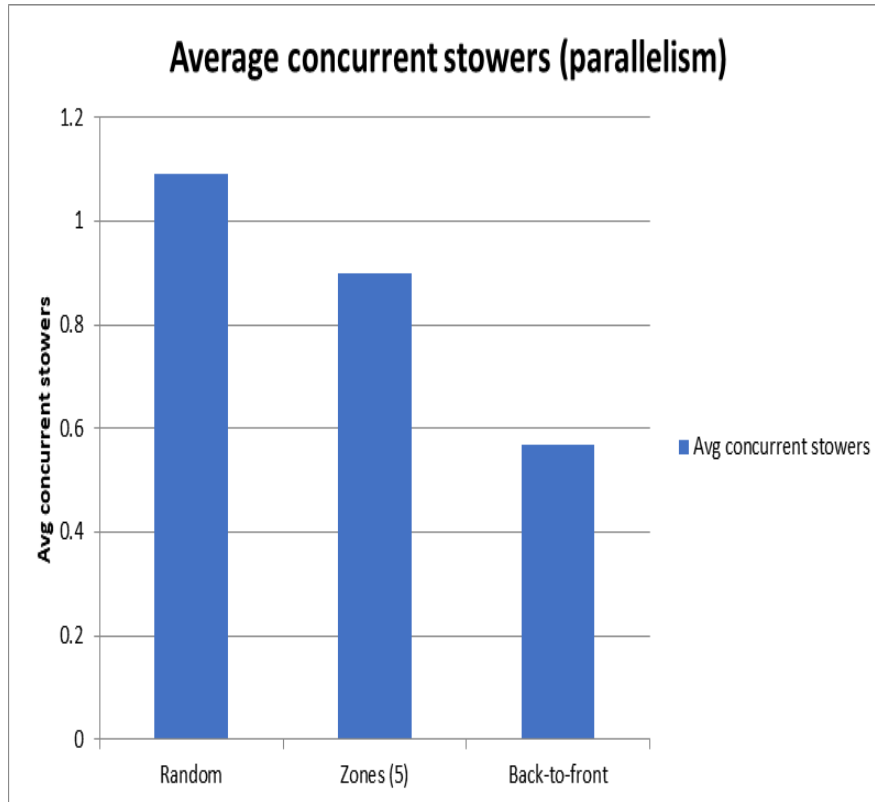


Figure 3. Average parallelism (concurrent stowing) by strategy.

**Exact theory.** Implementing the recurrence of Section 2.4 gives the same value as the simulation, **bit-for-bit**:

| Case                     | T (simulation) | T (theory) | Match |
|--------------------------|----------------|------------|-------|
| Random, R=30, S=3        | 331            | 331        | OK    |
| Back-to-front, R=30, S=3 | 627            | 627        | OK    |
| Zones (5), R=30, S=3     | 420            | 420        | OK    |
| Random, R=30, S=6        | 466            | 466        | OK    |
| Back-to-front, R=30, S=6 | 1080           | 1080       | OK    |

A brute-force sweep over  $R = 1..8$ ,  $S = 1..5$ , four boarding patterns and up to 6 seats per row (25,600 cases) gave **0 mismatches**. The recurrence is therefore an exact theory of the process, not an approximation.

**Clustering and the critical path.** Define the *clustering*  $\kappa(\pi)$  of a boarding order as the number of adjacent passenger pairs that share the same row (back-to-front maximizes  $\kappa$ ; random nearly minimizes it). In the recurrence, consecutive same-row passengers create a chain of  $+S$  weights at a single cell ( $r_j = c$ ), which lengthens the longest path — so  $T$  increases with clustering, consistent with all simulations. The longest-path view thus makes the mechanism explicit: **clustering stacks stowing work on one cell and lengthens the critical path.**

## 4. Discussion

Random boarding is fastest because it spreads passengers across rows so several rows stow simultaneously (highest parallelism, least blocking). Back-to-front is slowest because its six same-row passengers board consecutively and must stow at one cell one by one — in the theoretical model, a long chain of +S weights on each cell. The sensitivity analysis (Figure 2) confirms the gap grows with stow time  $S$ , exactly as the recurrence's +S blocking term predicts.

## 5. Limitations and Future Work

- Simplified model: no assigned-seat interference, no dual aisle or second door.
- The theoretical recurrence uses a naive  $O(N)$  search for  $j$ ; it can be reduced to  $O(N \cdot R)$  overall with preprocessing.
- Future work: prove monotonicity of  $T$  in clustering; add the "Steffen method" and window-middle-aisle boarding; test a model in which parallelism is disabled as a direct causal test.

## 6. Conclusion

Dispersed (random) boarding outperforms clustered (back-to-front) boarding. We explain this with a mechanism metric (parallelism) corroborated by an **exact longest-path (max-plus) model** that computes the boarding time without simulation and matches it bit-for-bit. Clustering same-row passengers stacks stowing weights on a single aisle cell and lengthens the critical path — the underlying principle behind the observed gap.

## References

Steffen, J. H. (2008). Optimal boarding method for airline passengers. *Journal of Air Transport Management*. Van Landeghem, H., & Beuselinck, A. (2002). Reducing passenger boarding time in airplanes. *European Journal of Operational Research*. Bachmat, E., et al. (2006). Analysis of airplane boarding via space-time geometry and random matrix theory.